Greedy Knapsack problem

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Course: Design Analysis and Algorithm
Code: BCSE204P
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Code:

```
#include<stdio.h>
int main()
{
	int n;
	printf("Name : Mistry Nishk Nileshkumar");
	printf("\nRegistration Number: 24BCE5236");
	printf("\nEnter number of item: ");
	scanf("%d", &n);
	int capacity, w[n], p[n];
	float r[n], mv=0;
	printf("Enter weight: ");
	for(int i=0; i<n; i++){
		scanf("%d", &w[i]);
	}
	printf("Enter profit: ");
	for(int i=0; i<n; i++){
		scanf("%d", &p[i]);
	}
	printf("Enter total capacity: ");
	scanf("%d", &capacity);
	for(int i=0; i<n; i++){
		r[i] = (float)p[i] / w[i];
	}
	for(int i=0; i<n; i++){
		for(int j=0; j<n-i-1; j++){
			if(r[j] < r[j+1]){
				float a = r[j];
				r[j] = r[j+1];
				r[j+1] = a;
				int wx = w[j];
				w[j] = w[j+1];
				w[j+1] = wx;
				int px = p[j];
				p[j] = p[j+1];
				p[j+1] = px;
			}
		}
	}
	printf("Table: \n");
	printf("weight\tprofit\tratio\n");
	for(int i=0; i<n; i++){
		printf("%d\t%d\t%.2f\n", w[i], p[i], r[i]);
```

```
        }
        for(int i=0; i<n; i++){
                if(capacity>=w[i]){
                        capacity = capacity - w[i];
                        mv = mv + p[i];
                        // printf("%d\n", capacity);
                }
                else if(w[i] > capacity){
                        float t = (float) (r[i]*capacity);
                        mv = mv + t;

                        capacity = capacity - capacity;
                }
                if(capacity==0){
                        //printf("\nKnapsack is full.");
                        break;
                }
        }
        printf("\nThe maximum profit possible: %.2f\n", mv);
        return 0;
}
```

Screenshots:

Terminal:

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```
Enter total capacity: 15
Table:
weight  profit  ratio
3       10      3.33
2       5       2.50
4       3       0.75
10      7       0.70
12
10
6

The maximum profit possible: 22.20
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/Nishk$ gcc Knapsack_problem.c -o Knapsack_problem
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/Nishk$ ./Knapsack_problem
Enter number of item: 4
Enter weight: 4
8
3
1
Enter profit: 6
9
8
5
Enter total capacity: 12
Table:
weight  profit  ratio
1       5       5.00
3       8       2.67
4       6       1.50
8       9       1.12

The maximum profit possible: 23.50
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/Nishk$ gcc Knapsack_problem.c -o Knapsack_problem
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/Nishk$ ./Knapsack_problem
Name : Mistry Nishk Nileshkumar
Registration Number: 24BCE5236
Enter number of item: 4
Enter weight: 10
3
4
2
Enter profit: 7
10
3
5
Enter total capacity: 15
Table:
weight  profit  ratio
3       10      3.33
2       5       2.50
4       3       0.75
10      7       0.70

The maximum profit possible: 22.20
student@admin205-ThinkCentre-neo-50s-Gen-5:~/Documents/Nishk$
```

Text Editor:

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Activities | Text Editor | Dec 16 17:25

Open | Knapsack_problem.c ~/Documents/Nishk | Save

Knapsack_problem | Knapsack_problem.c

```c
#include<stdio.h>
int main()
{
	int n;
	printf("Name : Mistry Nishk Nileshkumar");
	printf("\nRegistration Number: 24BCE5236");
	printf("\nEnter number of item: ");
	scanf("%d", &n);
	int capacity, w[n], p[n];
	float r[n], mv=0;
	printf("Enter weight: ");
	for(int i=0; i<n; i++){
		scanf("%d", &w[i]);
	}
	printf("Enter profit: ");
	for(int i=0; i<n; i++){
		scanf("%d", &p[i]);
	}
	printf("Enter total capacity: ");
	scanf("%d", &capacity);
	for(int i=0; i<n; i++){
		r[i] = (float)p[i] / w[i];
	}
	for(int i=0; i<n; i++){
		for(int j=0; j<n-i-1; j++){
			if(r[j] < r[j+1]){
				float a = r[j];
				r[j] = r[j+1];
				r[j+1] = a;
				int wx = w[j];
				w[j] = w[j+1];
				w[j+1] = wx;
				int px = p[j];
				p[j] = p[j+1];
				p[j+1] = px;
			}
		}
	}
	printf("Table: \n");
	printf("weight\tprofit\tratio\n");
	for(int i=0; i<n; i++){
		printf("%d\t%d\t%.2f\n", w[i], p[i], r[i]);
	}
	for(int i=0; i<n; i++){
		if(capacity>=w[i]){
			capacity = capacity - w[i];
			mv = mv + p[i];
			// printf("%d|n", capacity);
		}
		else if(w[i] > capacity){
			float t = (float) (r[i]*capacity);
			mv = mv + t;
```

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Knapsack_problem | Knapsack_problem.c

```c
	for(int i=0; i<n; i++){
		scanf("%d", &w[i]);
	}
	printf("Enter profit: ");
	for(int i=0; i<n; i++){
		scanf("%d", &p[i]);
	}
	printf("Enter total capacity: ");
	scanf("%d", &capacity);
	for(int i=0; i<n; i++){
		r[i] = (float)p[i] / w[i];
	}
	for(int i=0; i<n; i++){
		for(int j=0; j<n-i-1; j++){
			if(r[j] < r[j+1]){
				float a = r[j];
				r[j] = r[j+1];
				r[j+1] = a;
				int wx = w[j];
				w[j] = w[j+1];
				w[j+1] = wx;
				int px = p[j];
				p[j] = p[j+1];
				p[j+1] = px;
			}
		}
	}
	printf("Table: \n");
	printf("weight\tprofit\tratio\n");
	for(int i=0; i<n; i++){
		printf("%d\t%d\t%.2f\n", w[i], p[i], r[i]);
	}
	for(int i=0; i<n; i++){
		if(capacity>=w[i]){
			capacity = capacity - w[i];
			mv = mv + p[i];
			// printf("%d|n", capacity);
		}
		else if(w[i] > capacity){
			float t = (float) (r[i]*capacity);
			mv = mv + t;

			capacity = capacity - capacity;
		}
		if(capacity==0){
			//printf("\nKnapsack is full.");
			break;
		}
	}
	printf("\nThe maximum profit possible: %.2f\n", mv);
	return 0;
}
```

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